A Maximal-Monotone Stick–Slip Atlas for the Coulomb Oscillator

HCS Research Program

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Abstract

We make the scalar Coulomb oscillator unambiguous at zero velocity by freezing the maximal-monotone Sign graph together with a viability rule. For $\dot{x} = v$, $\dot{v} = -\omega^2 x - c\xi$, the static interval is $|x| \le a_f = c/\omega^2$. Each slip branch is a shifted harmonic oscillator, giving $\dot{E} = -c|v|$, an exact rest half-cycle map, an integer finite-capture formula, and closed first-turn phases for arbitrary signed velocity. The $c = 0$ harmonic face is separated because it has no capture. This is a source-local nonsmooth-mechanics theorem: it supplies no arithmetic owner, target determinant, or Hilbert–Pólya claim.

1 Inclusion and selection law

Write

$$\dot{x} = v, \qquad \dot{v} = -\omega^2 x - c\xi, \qquad \xi \in \mathrm{Sign}(v), \quad \omega > 0,\ c \ge 0, \tag{1}$$

where $\mathrm{Sign}(0) = [-1, 1]$. Put $a_f = c/\omega^2$. At $v = 0$, the selected viability law sticks exactly when $|x| \le a_f$; an exterior rest releases inward. This convention is part of the model and removes the otherwise ambiguous zero-velocity branch.

Theorem 1 (Global trajectory and energy)**.** *The selected maximal-monotone trajectory exists uniquely for all forward and backward finite times. On every slip interval,*

$$E = \tfrac{1}{2}(v^2 + \omega^2 x^2), \qquad \dot{E} = -c|v|, \tag{2}$$

while on a sticking interval $\dot{E} = 0$.

Proof. For $v > 0$, (1) is the smooth equation $\ddot{x} + \omega^2 x + c = 0$; for $v < 0$ it is $\ddot{x} + \omega^2 x - c = 0$. The Sign graph is maximal monotone, and the viability selection keeps a rest in the closed static interval. Standard one-dimensional maximal-monotone uniqueness then concatenates the smooth slip arcs and the sticking intervals without a Zeno ambiguity. Multiplying the slip equation by v gives (2). □

2 Rest starts and exact capture

For $x(0) = A > a_f$, $v(0) = 0$, the first motion has $v < 0$, center $+a_f$, and

$$x(t) = a_f + (A - a_f)\cos(\omega t), \qquad x_1 = 2a_f - A. \tag{3}$$

If A_k is the signed turning point after the k-th complete moving half-cycle, then

$$A_k = (-1)^k (A - 2k a_f), \qquad |A_{k+1}| = |A_k| - 2a_f. \tag{4}$$

Consequently the exact number of moving half-cycles for $A > a_f$ is

$$n = \left\lceil \frac{A - a_f}{2a_f} \right\rceil, \tag{5}$$

and the stopping turn A_n lies in $[-a_f, a_f]$. A rest already in that interval has $n = 0$. Negative rests follow by reflection.

Proposition 1 (Work check)**.** *The first-turn energy drop from a positive rest is $E(A, 0) - E(2a_f - A, 0) = c\,|A - (2a_f - A)|$, exactly the friction work.*

3 Arbitrary velocity: a partial first arc

Assume $c > 0$. If $v_0 > 0$, the first branch has center $-a_f$, radius

$$R = \sqrt{(x_0 + a_f)^2 + (v_0/\omega)^2},$$

phase $\theta_0 = \operatorname{atan2}(-v_0/\omega, x_0 + a_f)$, first-turn time $\tau_0 = -\theta_0/\omega$, and next turn $X_1 = -a_f + R$. If $|X_1| > a_f$, the remaining complete half-cycle count is

$$r = \left\lceil \frac{|X_1| - a_f}{2a_f} \right\rceil, \qquad \tau_{\text{stop}} = \tau_0 + r\pi/\omega. \tag{6}$$

If $|X_1| \le a_f$, the first arc itself stops in the static interval.

For $v_0 < 0$, replace the center by $+a_f$, use

$$R = \sqrt{(x_0 - a_f)^2 + (v_0/\omega)^2}, \quad \theta_0 = \operatorname{atan2}(-v_0/\omega, x_0 - a_f),$$

with first-turn time $(\pi - \theta_0)/\omega$ and next turn $X_1 = a_f - R$. The same ceiling rule applies. The first segment here is a partial slip arc, not a complete half-cycle; only r is an integer count of later complete half-cycles. At an exterior rest, the positive side has center $+a_f$, phase zero, and time π/ω; the negative side has center $-a_f$, phase π, and the same time.

References

[1] J.-J. Moreau, Application of convex analysis to some problems of dry friction, in *Trends in Applications of Pure Mathematics to Mechanics*, Vol. 2, Pitman (1979), 263–280.

[2] B. Brogliato, *Nonsmooth Mechanics*, Springer (1999).

[3] J. P. LaSalle, Some extensions of Liapunov's second method, *IRE Transactions on Circuit Theory* 7 (1960).

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; no target arithmetic, target-zero, determinant-matching, or Hilbert–Pólya claim. **Data and code.** All rows and audits accompany HCS-C238. This is not external peer review. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checks displayed formulas and metadata.